

Macro Threshold & Regime Experiments: Pipeline Updates and Review Answers

This note pulls together what we actually shipped on the macro intelligence pipeline between the 2026-06-06 regime v2 experiment run and the 2026-06-09 nightly briefing fixes. It is meant for Rohit and Divyanshu as a review packet: plan status vs reality, answers to the FM/regime questions, and the friction we hit along the way.

Where the forward plan stands today

The forward plan (`MACRO_TH_EXP_PLAN.md`) said we are past shadow experiments and into Rohit review plus wiring decisions. That is still accurate.

Plan phase	Planned state	Actual state (2026-06-09)
Shadow experiments A–H	Done	Done (2026-06-06 run)
Rohit review packet	P0 item #1	Done (<code>MACRO_TH_EXP_STATUS_ANALYSIS.md</code> §3)
CONFIG B4 window fix	P1, not started	Open (HY/VIX/VXTS still <code>full</code> , WALCL now <code>full</code> in production nightly but B4 audit not re-run)
Combo B confirmed-only slice	P0 #3	Open (DB still has 89 B WATCH rows, 0 ACTIVE)
Production v2 regime swap	Post sign-off	Not wired (legacy labels in nightly PDF)
HMM production	Dec 2026+	Deferred (prototype only)

Worth noting here: the **production nightly briefing pipeline** did get a separate round of fixes (Divyanshu's 11-point report review under `testing/macro_report_updates/`). That work is live in engine code and PDF output. The **regime v2 shadow pipeline** is complete in analysis mode but not swapped into production yet.

Pipeline updates we made

1. Regime v2 experiment suite (shadow / research)

Single entry point:

```
.venv/bin/python scripts/run_regime_v2_experiment_suite.py
```

Deliverable	What the pipeline does now	Production wired?
A	5-dimension shadow regime backfill (1,901 Fridays)	No
B	TWY_ROC + dual percentiles (14,457 rows, 0 fallbacks)	TWY_ROC not in nightly pull
C	Emission vectors (8,805 daily rows)	No daily job
D	HMM prototype (k-means style)	Deferred
E	Combo C cancel probability function	Built, not on briefing
F	F2/F2a quant defs + F4 steepening grid	Shadow only
G	7WK grind + VIX suppressed persistence	Not in briefing framing
H	298-combo discovery (132 survivors, 62 promotion candidates)	Monthly re-run possible, no new named combos

Artifacts land in `macro_intelligence/analysis/regime_v2_experiments/` including `X-FM_all.json` and `X_COMBO_regime_slices.json`.

2. Production nightly briefing pipeline (Divyanshu review)

These changes affect what you see in `runic_briefing_*.pdf` today:

Update	Before	After
Combo hit rates	Uniform 3M, bullish-only SQL	Per-combo horizon + bearish direction (E = 12M bearish)
Combo C terminal state	INACTIVE after cancel	CANCELLED + <code>cancel_date</code>
Combo F duration	Week count only	Week count + <code>started YYYY-MM-DD</code>

Update	Before	After
WALCL percentile	Rolling 3Y (inflated ~85th on flat MoM)	Full history from 2008
Combo B HY gate	Single absolute floor	Dual: 400bps + 80th pctile
Posture label	BRAVE	EASY MONEY
Combo G hit rate	Shown like other combos	N/A (timing warning only)
PDF combo table	Fixed narrow columns, text overlap	Paragraph wrap + full-width columns

Tests: 15 combo/metadata tests passed; 5 briefing renderer tests passed after PDF table fix.

Follow-up scripts: `scripts/analyze_combo_g_b_cascade.py` (G→B cascade + HY audit), MRU-01..03 executed.

Answers to Rohit's FM and regime questions

Source ask: `additional_details.md`. Backtest JSON: `X-FM_all.json`, `X_COMBO_regime_slices.json` (1990–2026, shadow v2 regime tags).

Extreme short FM (<15th pctile): contrary indicator?

Rohit claimed ~87.5% SPX higher 3m after extreme short FM (7/8 Combo B cases).

Test	n	SPX up 3m	Verdict
Combo B fires (all DB rows incl. WATCH)	89	79.8%	Mostly validated, lower rate, much larger n
Raw FM <15 crossings	35	60.0%	Weaker than Rohit's headline number
FM "wrong" (= SPX up) at extreme short	35	60%	Direction OK, magnitude below 87.5%

Regime isolation (raw FM <15, SPX up 3m by fed_cycle_v2):

Regime	n	SPX up 3m
EASING	22	54.5%

Regime	n	SPX up 3m
TIGHTENING	7	57.1%
EASY	6	83.3%
FLAT curve	6	33.3%
INVERTED curve	4	75.0%
NORMAL curve	22	63.6%

The contrary edge is **conditional**. It holds best when Combo B legs align (VIX + HY + CFTC together), not on FM percentile alone. Flat-curve regimes break the pattern.

Extreme long FM (>85th pctile): Combo D territory?

Horizon	Raw FM >85: SPX down	Raw FM >85: FM wrong (SPX up)
1 week	41%	59%
3 months	18%	82%

Horizon	Combo D fires: SPX down	Combo D fires: FM wrong
1 week	38.5%	61.5%
3 months	28.1%	71.9%

Rohit is **partially right**. Short-horizon wrong rates (~59–62%) sit below his 72–85% band. At 3m, FM extreme long is wrong most of the time because SPX keeps grinding up. Combo D is regime-dependent: only 18.3% SPX down 3m in HIKING_LATE (n=197) vs 43.2% in CUTTING_LATE (n=155).

Moderate FM (25th–75th): "trend is your friend"?

Metric	Result
Crossings	84
SPX up 3m	76.2%
Avg 3m return	+3.15%

Rohit flagged this as not necessarily accurate. The backtest shows bullish 3m outcomes most of the time, but that looks a lot like **baseline equity drift**, not a standalone alpha signal. No actionable edge fading moderate FM.

Regime impact on named combos A–G (3m horizon)

Combo	n	Overall 3m "hit"	Strongest slice	Weakest slice
A (bearish)	174	23% SPX down	CUTTING_LATE 50% (n=26)	QE 20% (n=112)
B (bullish)	89	79.8% SPX up	HIKING_LATE 83% (n=48)	CUTTING_LATE 76% (n=41)
C	4	0% up	n too small	n/a
D (bearish)	452	28% SPX down	CUTTING_LATE 43%	HIKING_LATE 18%
E (bearish)	507	20% SPX down	(flat across regimes)	n/a
F (bullish)	704	74.9% SPX up	QE 82% (n=212)	CUTTING_LATE 64% (n=248)
G	0	n/a	No fires in DB	n/a

Combo B is robust across fed cycles. Combo D only really works outside HIKING_LATE. Combo F gets a boost in QE, which supports the Part H beta filter idea.

Plan §6 open questions: closure table

Matches MACRO_TH_EXP_STATUS_ANALYSIS.md §5 against MACRO_TH_EXP_PLAN.md immediate questions.

#	Question (consolidated plan)	Answer / status
1	TWY_ROC ±0.30pp bands	Validated starting point; Apr 2025 anchor passes (−0.55pp DOVISH)
2	F4 trough −50 vs −80, steep +15 vs +40	Grid run; win rates <35%; mechanism gate only
3	Apr 2025 DGS2 vs fed_cycle	Divergence confirmed; TWY_ROC adds leading signal

#	Question (consolidated plan)	Answer / status
4	Dual percentile <50 fallback	Implemented; 0 fallbacks in backfill
5	Beta 55% vs 60%	Both in JSON; Rohit decision pending
6	2-of-3 vs 3-of-3 legs	Diagnostic only; no production change
7	6mo before HMM prod	DEFER ; live C1 vectors not wired
8	T10Y2Y vs Ahil steepening work	F2/F2a in shadow; Ahil review pending
9	Classifier prompt update	Pending Rohit sign-off
10	Rohit FM Q&A	Answered above ; partial validation

Questions queued for Rohit sync (from forward plan §2)

#	Question	Recommended answer from data
1	Approve fed_cycle_v2 4-state (PIVOTING n=27)?	Merge PIVOTING into EASING or drop until $n \geq 30$
2	F4 steepening-short stays mechanism-only?	Yes ; grid n too small for statistical promotion
3	Beta bar 55% or 60%?	Need per-combo call; 62 candidates waiting
4	TWY_ROC ± 0.30 pp for classifier?	Yes as starting point (Apr 2025 pass)
5	Start daily emission_vectors job?	Yes if C approved ; starts 6-month HMM clock

Challenges we faced (honest list)

Challenge	What happened	Impact
Shadow vs production split	v2 regime labels live only in <code>macro_regime_log_v2</code> , not nightly PDF	Rohit sees legacy CUTTING_LATE etc. in briefing while analysis uses v2
CONFIG B4 window mismatch	HY/VIX/VXTS on <code>full</code> , plan wanted <code>rolling_3y</code> ; WALCL was wrong until macro report fix	Percentile ranks differ between experiment audit and production nightly

89 WATCH rows vs Rohit's "8 confirmed"

Challenge	What happened	Impact
Combo B count confusion		Hit rate backtests mix partial-leg WATCH with true capitulation
FM report terminology	Bearish signals sometimes labeled "SPX down" when JSON stores FM wrong rate	Audit requires reading raw JSON, not summary tables
PIVOTING sample size	n=27 below ≥ 30 obs gate	A1 validation fails; blocks clean 4-state promotion
HMM blocked on live data	Need 6 months production emission vectors	Earliest Dec 2026; prototype shows no Sharpe lift
FRED API timeouts	Full nightly pull hit HTTP 504 on 2026-06-09	Had to fall back to DB cache for some runs
PDF table overlap	ReportLab fixed columns + plain strings	Duration text bled into Direction column; fixed with Paragraph wrap
Zero Combo G fires	0 G in DB and 2007–2026 rescan	Cannot validate G→B 3–4 week lead from automated data
HY dual gate vs Oct 2022	427 bps passes abs floor, fails 80th pctile on full history	Tension between new rule and canonical B date (MRU-02)
Part H Claude story step	Ran with <code>use_claude=False</code>	62 promotion candidates have no economic narrative yet
CFTC pre-2010 sparsity	FM bands thin before TFF era	Long-history FM stats dominated by post-2010 data

On the flip side, the experiment suite itself re-runs in about 2 minutes, which made iteration tolerable. The painful parts were mostly definition mismatches (WATCH vs ACTIVE, SPX down vs FM wrong) rather than runtime.

What I'd change based on this

1. **Re-slice Combo B as confirmed-only (3/3 legs ACTIVE)** before the next Rohit sync. That single filter probably closes most of the gap between 79.8% and Rohit's 87.5%.
2. **Fix CONFIG B4 windows and re-run the suite** so shadow and production percentiles speak the same language.

3. **Wire cancel probability to the briefing** (Part E is built; dashboard display is not).
 4. **Hold HMM and new named combos** until Rohit signs A/B/C and we have live vectors.
 5. **Add regime footnotes to combo hit rates** in the PDF (fed_cycle slices from `X_COMBO_regime_slices.json`) once v2 labels swap in.
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My doubts and questions

- Is PIVOTING a real economic state or just label noise at n=27? Merging it into EASING feels pragmatic but might hide genuine pivot weeks.
 - The moderate FM "76% SPX up" result: I have not stripped out unconditional market drift. Does it still look special after subtracting a simple buy-and-hold baseline?
 - Combo C has n=4 in the regime slice table. Should we even show C hit rates in the briefing until we have more cancelled/completed episodes?
 - Part H found 62 combos at $\geq 80\%$ hit rate. How many are just re-labelling existing A–G leg combinations vs genuinely new structure?
 - Ahil's steepening-of-inversion work vs our F4 grid: are we measuring the same thing, or will that review surface a definitional mismatch?
 - For production GO-live, what is the rollback plan if v2 fed_cycle labels confuse readers who have been trained on CUTTING_LATE / HIKING_LATE for months?
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Report prepared 2026-06-09. Sources: `MACRO_TH_EXP_STATUS_ANALYSIS.md`, `MACRO_TH_EXP_PLAN.md`, `additional_details.md`, `macro_intelligence/analysis/regime_v2_experiments/`, `testing/macro_report_updates/MACRO_REPORT_UPDATE_STATUS.md`.